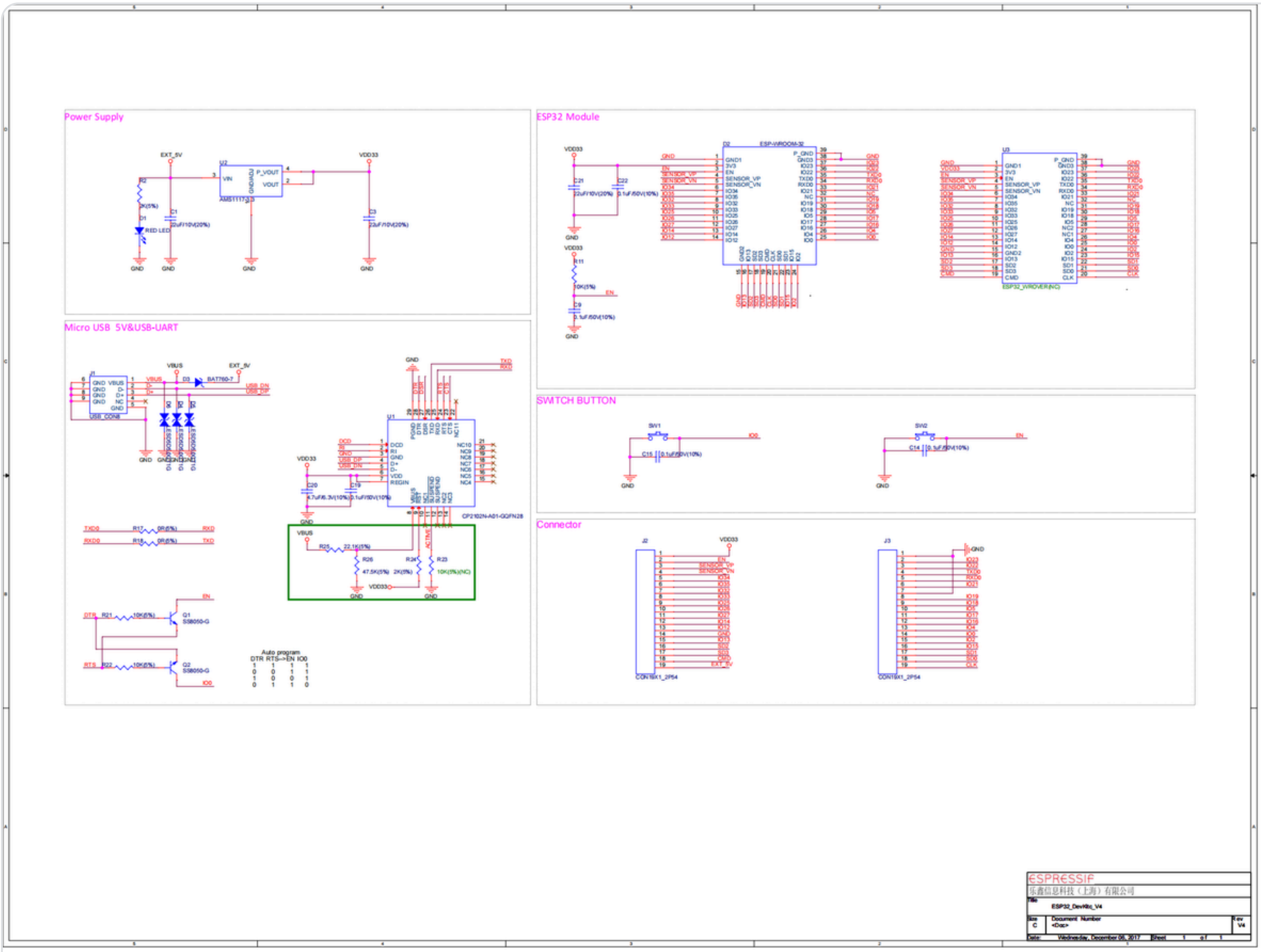
Debugging Tutorial for AmazingHand (PWM Analog Servo, i.e., No Real Servo)

⚠️ **AmazingHand with real servos can skip this tutorial!**

ESP32 Development Board Circuit Diagram



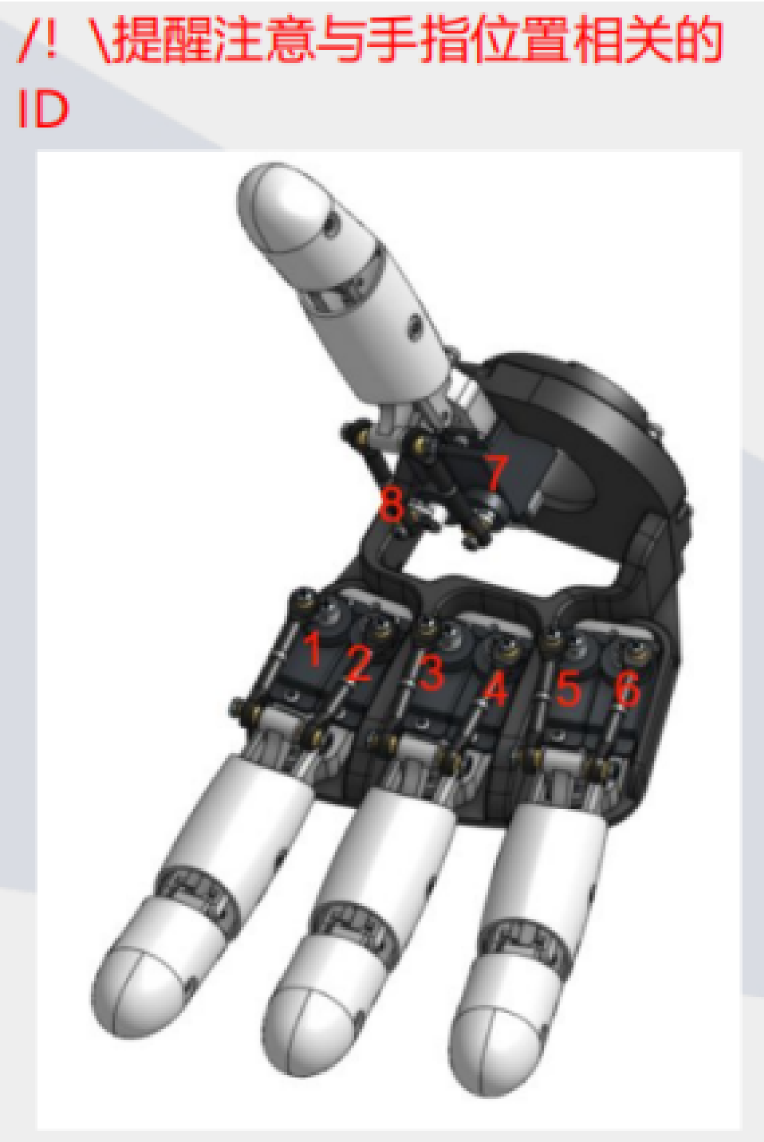
esp32_devkitc_v4-sch.pdf

Currently, there is only one way to debug the PWM servo dexterous hand, which is through a single-chip microcontroller. Here, we will use the ESP32 for illustration.

First, download the " Dexterous Hand Debugging.zip " Compressed Packet and extract it to ensure that the system has installed the arduio software

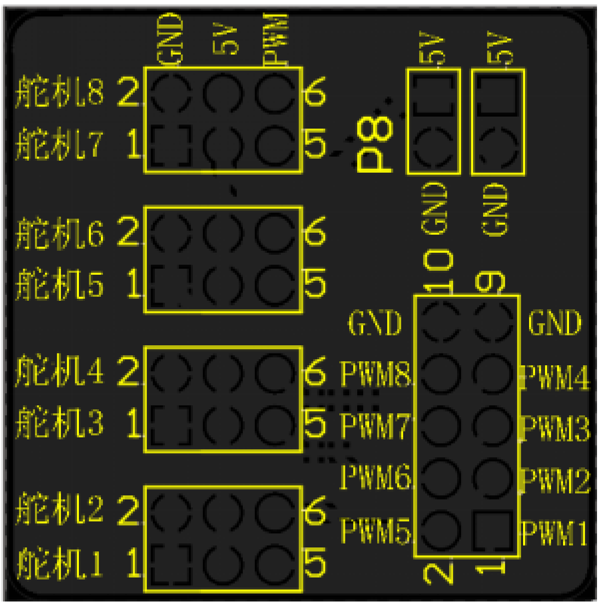
1. Determine the Servo ID

Although PWM servos do not require setting a servo ID like TTL servos do, they still need to have a logical ID clearly defined, and the corresponding relationship is as follows:

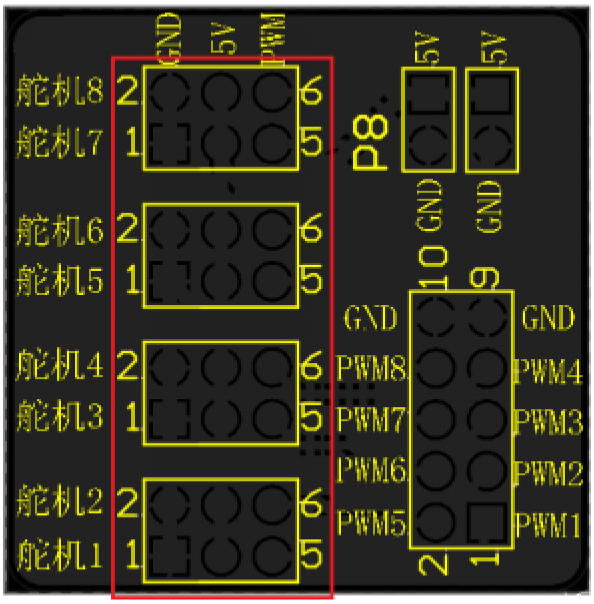


右手ID顺序

2. Wiring Instructions



1. Servo <--> Adapter Board: Insert the 3P plugs of the 8 servos into the pin headers of Servo 1 - Servo 8 according to the determined ID numbers.



2. Adapter Board <--> ESP Development Board:

(1) There are two sets of 5V GND power supply ports. One set is led out by a TPYE-C cable and connected to a 5V-3A power supply. The other set can be led out to supply power to the development board (5V pin), so that during operation, the ESP32 no longer needs to be powered through the Micro port.

(2) PW1 - PW8: Pulse control interfaces for 8-channel motors. Connect them in the following order

For example: SERVO_PIN1 is PWM1, and PWM1 of the adapter board is connected to IO23 of the development board

Amazing_Hand_Demo

```
#define SERVO_PIN1 23
#define SERVO_PIN2 22
#define SERVO_PIN3 21
#define SERVO_PIN4 19
#define SERVO_PIN5 18
#define SERVO_PIN6 5
#define SERVO_PIN7 17
#define SERVO_PIN8 16
```

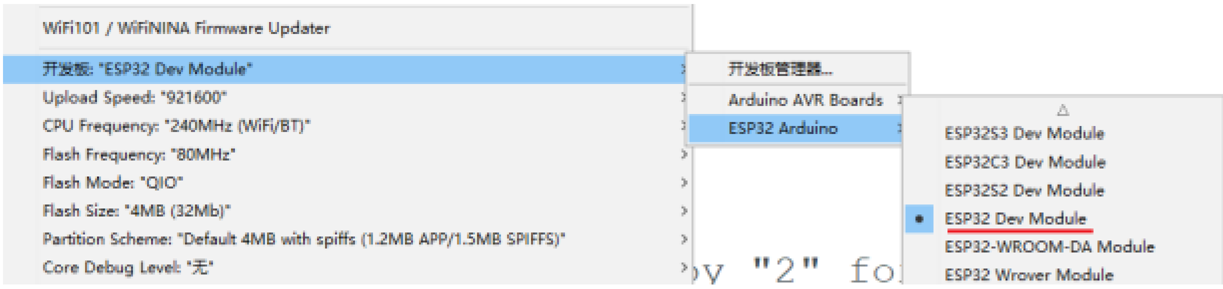
(3) GND: At least one must be connected for power supply .

3. Fix the servo horn

1. Download the program in "Used When Installing White Servo Horns"

Function of this program: Position the gear of the servo motor at the center (90 degrees), and subsequent action angles are all based on this central position.

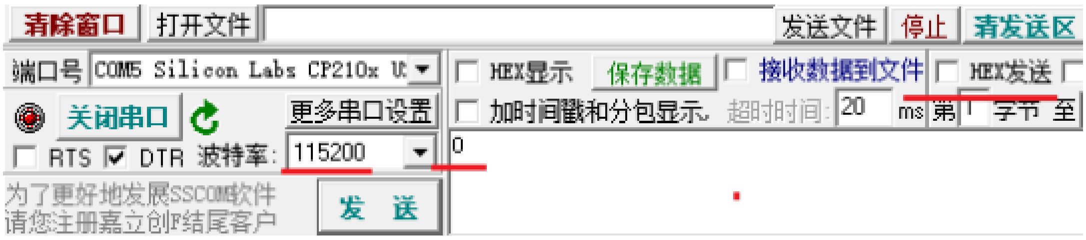
(1) Select "ESP32 Dev Module" for the development board type



(2) According to the servo number to be debugged (1-8), modify the following position. The value below controls Servo 1.

```
if (Serial.available())
{
  int angle = Serial.parseInt();
  if (angle >= 0 && angle <= 180)
  {
    setServoAngle(1, angle);
    Serial.print("Angle set to: ");
    Serial.println(angle);
  }
```

(3) After the program runs, enter the character "0" in the serial port assistant of the host computer, click "Send", and at this time the servo rotates to 0 degrees



(4) Install the test rudder arm vertically. For servos 1, 3, 5, and 7, the 0-degree position is at the bottom; for servos 2, 4, 6, and 8, the 0-degree position is at the top.


1、3、5、7号舵机0度位置

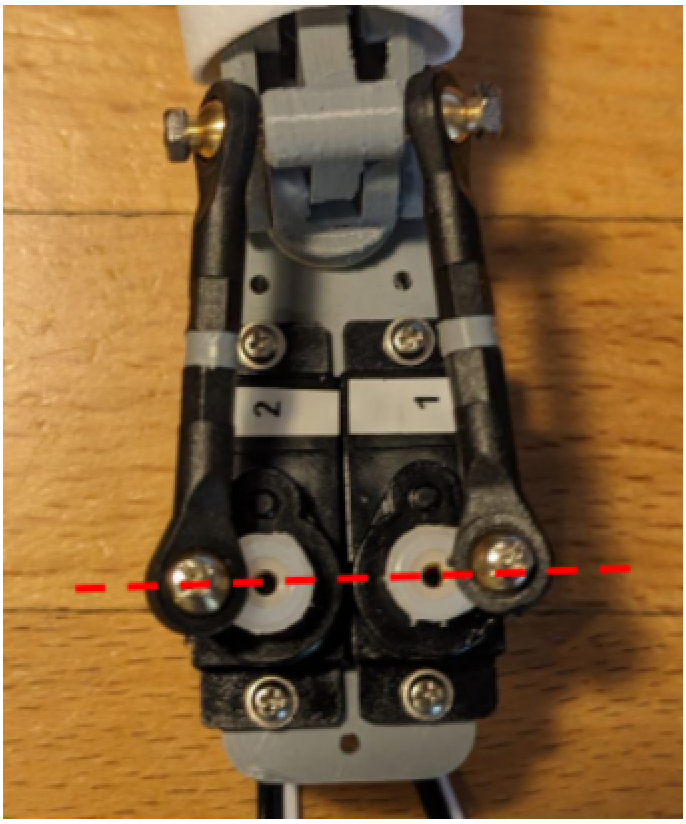

2、4、6、8号舵机0度位置

(5) Enter the character "90" in the serial port assistant, click "Send", and at this time the servo will rotate to 90 degrees. Observe whether the servo arm is perpendicular to the servo. If not, modify the value in midPulseWidth[], and the position of the value for servo No. 1 is as follows.

Re-download the program, click "Send" again, and repeat this process until the servo arm is perpendicular to the servo.

```
const int minPulseWidth[] = {500, 500, 500, 50
const int midPulseWidth[] = {1450, 1450, 1450,
const int maxPulseWidth[] = {2460, 2460, 2460,
void setup()
```

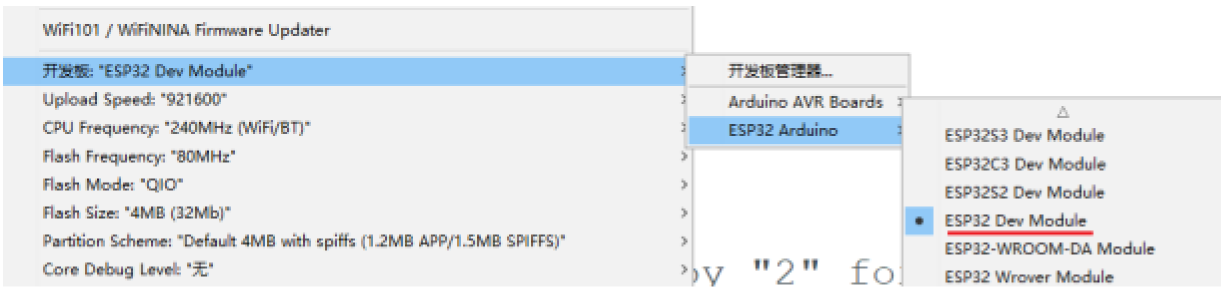
2. Install the servo horn on the gear, keeping the position as parallel as possible.



4. Run the "01 Demo Program"

Under the `Dexterous Hand Debugging \01 PWM Analog Servo \ Arduino Program (PWM Servo) \01 Demo Program` directory, find Amazing_Hand_Demo.ino and double-click to open it.

Select "ESP32 Dev Module" for the development board type



Compile and upload, and the dexterous hand will continuously run in a loop **"01 Demo Program"**